Daily Logic Drill #4

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Disproof by Counterexample · Proof by Contradiction.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition (≤15 s each)

Kill a universal claim with a single well-chosen example. Fluency, not first contact.

1. Give a counterexample to: "Every prime number is odd."

2. Give a counterexample to: "$n^2 \geq n$ for every real number n."

3. Give a counterexample to: "If n is prime, then $n+2$ is prime."

4. Give a counterexample to: "Every multiple of 4 is a multiple of 8."

5. To prove "there is no largest even integer" by contradiction, what should you assume first?

6. True or false: "If assuming 'not P' leads to something impossible, P must be true."

7. Give a counterexample to: "$\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ for all $a, b \geq 0$."

8. Give a counterexample to: "$(x+y)^2 = x^2 + y^2$ for all real x, y."

9. True or false: "A single counterexample is enough to disprove a 'for all' claim."

10. Give a counterexample to: "$n! + 1$ is always prime."

Section B Manipulation Drills (≤50 s each)

Hunt systematically, and write your first two contradiction proofs.

1. Five envelopes contain the numbers $6, 9, 14, 21, 25$ respectively. Each is sealed with a wax stamp showing either a star or a circle: the envelopes containing $6, 14, 25$ are star-sealed, and the envelopes containing $9, 21$ are circle-sealed. Claim: "Every star-sealed envelope contains an even number." Which envelope's contents provide a counterexample?

2. Give a counterexample to: "If a and b are both irrational, then $a+b$ is irrational."

3. Give a counterexample to: "If a and b are both irrational, then ab is irrational."

4. Prove by contradiction: "There is no largest even integer."

5. Give a counterexample (using $n = 0$) to: "For every integer n, $n/n = 1$."

6. Give a counterexample (using $n = 1$) to: "Every positive integer has at least one prime factor."

7. Prove by contradiction: "There is no smallest positive rational number."

8. Give a counterexample to: "If p and q are distinct primes, then $p + q$ is composite."

9. Give a counterexample to: "For every positive integer $n \geq 1$, $2^n - 1$ is prime."

10. Prove by contradiction: "If n is a positive integer, n and $n + 1$ share no common factor greater than 1."

Section C Substitution & Structure (≤90 s each)

Find the smallest counterexample, not just any counterexample — and write longer contradiction proofs.

1. Consider the statement: "($*$) If a whole number n is 2 more or 4 more than a multiple of 5, then n is prime." How many counterexamples to ($*$) are there in the range $1 \leq n \leq 50$?

2. Find the smallest positive integer n for which $n^2 + n + 41$ is *not* prime.

3. Give a counterexample to: "$n^2 + 1$ is always either prime or a perfect square, for every positive integer n."

4. Prove by contradiction: "There is no integer solution to $x^2 - 3y^2 = 2$."

5. Find the smallest positive integer n for which $3^n + 2$ is *not* prime.

6. Give a counterexample to: "If x and y are both non-integers, then $x + y$ is a non-integer."

7. Prove by contradiction: "If a, b are positive integers and $a^2 + b^2$ is odd, then a and b do not have the same parity."

8. Give a counterexample to: "$n^2 - n + 17$ is prime for every positive integer n."

Section D Challenge (TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Consider: I. "$n = 41$ is a counterexample to '$n^2 + n + 41$ is always prime'." II. "$n = 40$ is the smallest counterexample to '$n^2 + n + 41$ is always prime'." III. "$n = 4$ is a counterexample to '$n^2 + n + 41$ is always prime'." Which of I, II, III are true?

A) none

B) I only

C) II only

D) III only

E) I and II only

F) I and III only

G) II and III only

H) I, II and III

2. Prove by contradiction: "There do not exist positive integers a, b with $a^2 - b^2 = 1$."

3. A student claims: "For every positive integer n, at least one of $n, n+2, n+4$ is divisible by 3." Determine, with proof, whether this claim is true.

4. Give a counterexample to: "If p is prime, then $2^p + 1$ is prime."

5. Statement $(*)$: "For every positive integer $n \geq 2$, there is a prime strictly between n and $2n$." (a) Verify $(*)$ for $n = 2, 3, 4, 5$ by direct computation. (b) Does this verification constitute a proof of $(*)$? Justify your answer.

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths